Options Pricing & Hedging

Implied Volatility Surface · American Option Pricing · Delta Hedging · Portfolio Risk

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Dashboard: https://options-pricing-hedging-uvtsfcmrgeofhuvhg6qg83.streamlit.app

GitHub: https://github.com/Mrinal-g/options-pricing-hedging

1. Executive Summary

This project implements a production-style options pricing system for GOOG covering the full quantitative workflow: live data ingestion, IV extraction, SVI vol surface calibration, three-model pricing, delta hedging, and portfolio risk analytics—deployed as a 6-tab Streamlit dashboard with 167 passing tests.

The CRR binomial tree achieves a **94.4% bid-ask hit rate** (MAE \$0.201) on 177 options across 5 expiries. Two innovations drive this: **(1)** American-aware IV extraction using CRR inversion for puts (hit rate 14% → 74.7%), and **(2)** weighted SVI calibration with a minimum weight floor that recovers from 12.1% to 81.1% during the April 1 tariff selloff. Dynamic delta hedging removes 99.8% of P&L variance; a stress matrix quantifies how vol mismatch and transaction costs erode this under realistic conditions.

2. Data & Filtering

Data is sourced live from Yahoo Finance: spot price, 1-year daily history, and the full option chain. Five expiries are selected spanning 7d to 350d. An 8-step liquidity filter removes illiquid, stale, and mispriced contracts:

Step	Filter	Rationale
1	Bid > \$0.05	Removes zero-bid options with no real market
2	Mid > \$0.05	Eliminates penny options dominated by noise
3	Ask > Bid	Removes locked/crossed markets
4	TTM > 0	Drops expired contracts still in the chain
5	Open interest > 0	Ensures at least one trade has occurred
6	Strike within ±15% of spot	Focuses on liquid, tradeable strikes
7	Spread ratio ≤ 50%	Filters wide-spread illiquid wings
8	Staleness check	Removes quotes older than the session

On April 3, 2026, this filtering retained **177 options** from ∼800 raw contracts. The ±15% moneyness band and 50% spread ratio filters are the most aggressive, removing deep OTM options where bid-ask spreads exceed \$3–5 and IV extraction becomes numerically unstable.

3. Implied Volatility Surface

3.1 American-Aware IV Extraction

BSM inversion for American puts inflates IV by attributing early exercise premium to volatility. This project dispatches: calls use BSM inversion, puts use CRR (N=200) inversion. This fix alone improved hit rate from **14% to 74.7%**.

3.2 SVI Calibration

Each expiry slice is fitted with Gatheral's SVI model in total variance space:

$$w(k) \;=\; a \;+\; b\Big[\,\rho\,(k-m) \;+\; \sqrt{(k-m)^2+\sigma^2}\,\Big] \tag{1}$$

$w = \sigma_{\text{iv}}^2 \cdot T$ (total variance), $k = \ln(K/S)$ (log-moneyness), $\rho \in (-0.999,\, -0.01)$ (equity left skew)

Fitting uses **inverse-spread weights** (weight = 1/spread) so liquid ATM options dominate. A **minimum weight floor at 20% of ATM weight** prevents illiquid wings from becoming invisible during stressed markets—without this, ρ flipped to +0.46 and hit rate collapsed to 12.1% on April 1 data. With the floor: **81.1%**.

3.3 Surface Interpolation

The fitted SVI points are interpolated via `LinearNDInterpolator` over the (log-moneyness, TTM) domain in total variance space, with `NearestNDInterpolator` fallback for extrapolation. Querying the surface for any (K, T) returns a smooth, arbitrage-aware IV that feeds all three pricing engines.

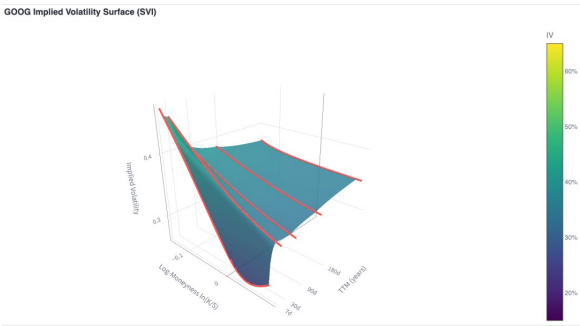

Figure 1: SVI implied volatility surface.

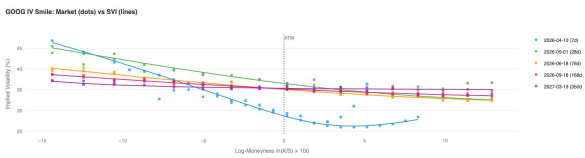

Figure 2: Market IV (dots) vs SVI fit (lines).

4. Pricing Models

4.1 Black-Scholes-Merton (European)

$$C = S\,N(d_1) - Ke^{-rT}N(d_2), \qquad P = Ke^{-rT}N(-d_2) - S\,N(-d_1) \tag{2}$$

$$d_1 = \frac{\ln(S/K) + (r+\sigma^2/2)\,T}{\sigma\sqrt{T}}, \qquad d_2 = d_1 - \sigma\sqrt{T} \tag{3}$$

BSM provides closed-form prices and all five Greeks $(\Delta, \Gamma, \nu, \Theta, \rho)$. It serves as the lower bound for American option prices—the gap is the early exercise premium (EEP). For GOOG calls (no dividend), EEP = 0 and BSM = CRR exactly, so BSM is only "wrong" for puts, where it systematically underprices by ignoring the option to exercise early. This makes BSM a useful diagnostic: the BSM–CRR gap isolates the EEP contribution.

4.2 CRR Binomial Tree (American)

$$u = e^{\sigma\sqrt{\Delta t}}, \quad d = 1/u, \quad p = \frac{e^{r\Delta t} - d}{u - d}, \quad \Delta t = T/N \tag{4}$$

$$V(i,j) = \max\Big\{ \underbrace{\text{payoff}(S_{i,j})}_{\text{exercise}},\ \underbrace{e^{-r\Delta t}\big[p \cdot V(i{+}1, j{+}1) + (1{-}p) \cdot V(i{+}1, j)\big]}_{\text{continuation}} \Big\} \tag{5}$$

CRR with N=200 steps is the production pricer for this project. At each node, the backward induction checks whether immediate exercise (the intrinsic value) exceeds the discounted continuation value. This is what makes CRR correct for American options: it prices the optionality of early exercise at every point in the tree. For European options, the max constraint is dropped and CRR converges to BSM as $N \to \infty$ (verified in the test suite with <\$0.01 difference at N=200). The tree also extracts the **early exercise boundary** $S^*(t)$—the critical spot below which exercising an American put immediately is optimal.

4.3 LSM Monte Carlo (American)

$$S(t{+}\Delta t) = S(t) \cdot \exp\Big[\big(r - \tfrac{\sigma^2}{2}\big)\Delta t + \sigma\sqrt{\Delta t} \cdot Z\Big], \quad Z \sim \mathcal{N}(0,1) \tag{6}$$

$$\mathbb{E}[\text{cont.} \mid S(t)] \approx \alpha_0 + \alpha_1 S(t) + \alpha_2 S(t)^2 \quad \text{(OLS on ITM paths; exercise if intrinsic} > \text{cont.)} \tag{7}$$

Longstaff-Schwartz simulates M=10,000 risk-neutral GBM paths with antithetic variates (variance reduction ∼30%). At each exercise date, only in-the-money paths are regressed—this is key to the algorithm's efficiency, as OTM paths contribute no exercise decision information. The polynomial basis $[1, S, S^2]$ is simple but sufficient for vanilla options; exotic payoffs might require higher-order or Laguerre bases. LSM introduces ±\$0.20–0.40 sampling noise at M=5,000 (the dashboard setting), making it unsuitable as the primary pricer. It serves as an independent cross-check: if LSM and CRR diverge by more than simulation noise, something is wrong.

5. Model Validation

5.1 Three-Date Progression

Each run exposed a failure mode leading to a targeted fix:

Date	Market	Hit Rate	MAE	Key Fix
Mar 28	Normal	74.7%	\$0.23	American-aware IV extraction
Apr 1	Tariff selloff	81.1%	\$0.30	Weighted SVI + weight floor
Apr 3	Post-selloff	**94.4%**	**\$0.201**	Validated on stable data

5.2 April 3 Results

Model	Hit Rate	MAE	Mean Error	MAPE
CRR (American)	**94.4%**	**\$0.201**	−\$0.016	**1.3%**
BSM (European)	85.9%	\$0.283	−\$0.207	1.6%
LSM (Monte Carlo)	62.7%	\$0.397	−\$0.345	2.2%

CRR prices 94.4% of options inside the bid-ask spread with near-zero systematic bias (mean error −\$0.016). The negative sign indicates a slight tendency to underprice, which is conservative from a market-making perspective. BSM shows a mean error of −\$0.207 on puts—the systematic underpricing expected when early exercise premium is ignored, confirming the EEP correction is working. LSM underperforms due to simulation noise at M=5,000 paths; the 62.7% hit rate would improve with more paths but at the cost of dashboard interactivity.

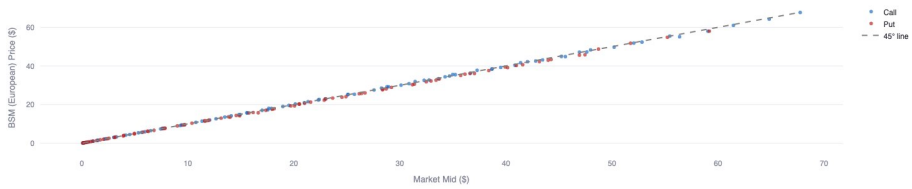

Figure 3: CRR model price vs market mid—calls (blue) and puts (red) tightly clustered along the 45° line across the full price range from \$0.50 to \$70.

5.3 Hit Rate by Moneyness

CRR achieves 100% hit rate on Deep ITM and ITM buckets, where bid-ask spreads are widest (\$0.50–2.00) and early exercise premium is most significant. ATM is the hardest bucket (86.8%) because spreads are tightest—the model must be accurate to within \$0.10–0.20 to land inside. OTM and Deep OTM perform well (92–93%) despite thin liquidity, validating the SVI surface's ability to price wings reliably.

Moneyness	CRR	BSM	LSM
Deep ITM	100.0%	100.0%	64.3%
ITM	100.0%	89.1%	84.8%
ATM	86.8%	78.9%	57.9%
OTM	92.0%	80.0%	48.0%
Deep OTM	93.3%	86.7%	53.3%

5.4 Volatility Risk Premium

The VRP compares ATM implied vol against historical realised vol over the matching horizon. A positive VRP means options are expensive relative to what actually materialised—the normal regime for equity markets, reflecting the insurance premium embedded in option prices.

Expiry	Days	ATM IV	Realised Vol	VRP (pp)
2026-04-10	7	29.1%	46.0%	**−16.9**
2026-05-01	28	36.9%	31.7%	+5.1
2026-06-18	76	35.0%	25.5%	+9.5
2026-09-18	168	35.2%	27.5%	+7.7
2027-03-19	350	35.4%	29.9%	+5.5

The 7-day expiry shows **−16.9pp**—the market underpriced near-term vol during the tariff selloff (realised 46% vs implied 29%). All other tenors show the typical equity VRP of +5 to +10pp, confirming that GOOG options are consistently priced above realised volatility.

All pricing, IV extraction, SVI calibration, hedging, and risk calculations are validated by **167**

unit tests across 6 test files, covering put-call parity, CRR–BSM convergence, IV round-trips, calendar no-arbitrage, and hedge P&L properties.

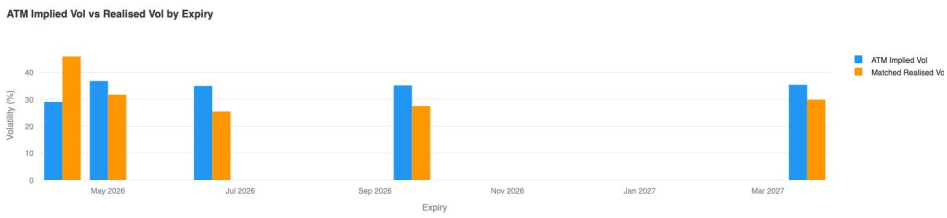

Figure 4: ATM implied vol vs matched realised vol by expiry.

6. Delta Hedging & Risk Analytics

6.1 Hedge Effectiveness

The hedge simulator sells an option and delta-hedges by trading the underlying stock daily. Three strategies are benchmarked on the same 200 simulated paths to isolate the effect of methodology, not path randomness:

Strategy	Std Dev	Var Reduction
Unhedged (sell option, no stock)	\$26.43	0%
Static (Δ_0 at t=0, never rebalance)	\$15.35	66.3%
Dynamic (daily rebalance to BSM Δ)	\$1.07	**99.8%**

The static hedge captures the initial delta exposure but cannot adapt as spot moves—hence only 66.3% variance reduction. Dynamic rebalancing tracks the changing delta, compressing P&L dispersion to \$1.07 (99.8% reduction). The residual comes from discrete hedging (daily, not continuous) and assumes realised vol equals implied vol—an assumption the stress matrix below relaxes.

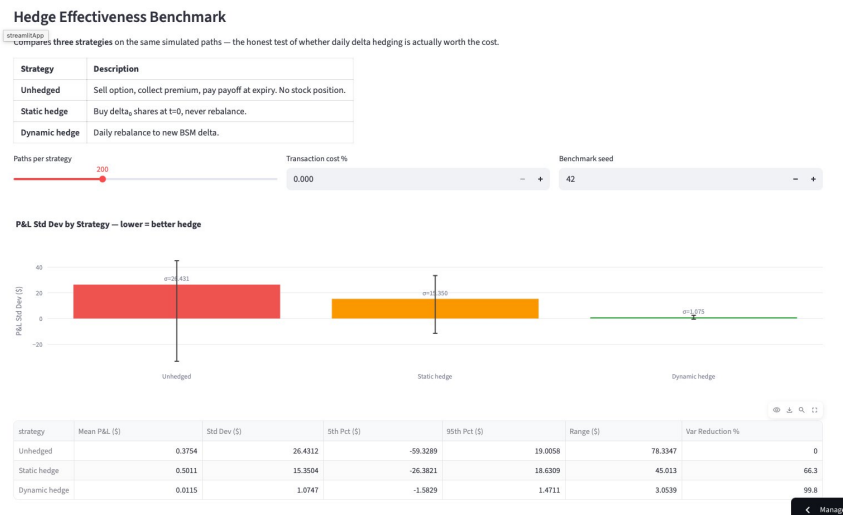

Figure 5: P&L standard deviation by strategy (200 paths). Dynamic hedging compresses dispersion to near-zero.

6.2 Hedge Stress Matrix

The 99.8% result assumes $\sigma_{\text{realised}} = \sigma_{\text{implied}}$ and zero transaction costs. The stress matrix shows mean P&L across 20 combined (realised vol × transaction cost) scenarios:

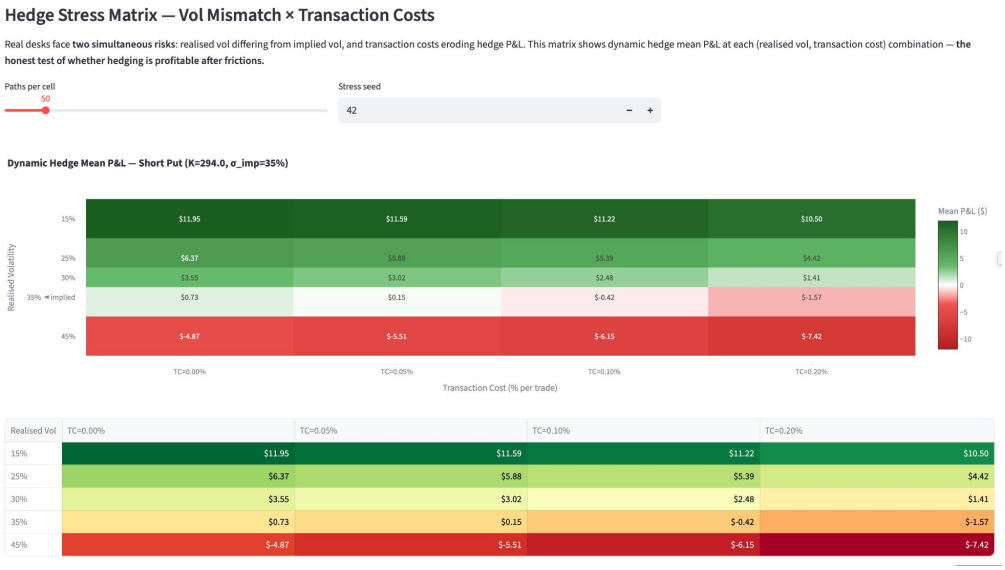

Figure 6: Hedge stress matrix—mean P&L across vol mismatch × transaction cost scenarios. Green = profitable (realised vol < implied). Red = losing (realised vol > implied). The 35% row marks implied vol. Heatmap and data table shown together.

At implied vol (35%), the hedge breaks even at zero cost ($0.73) and turns negative with >10bp costs. At 25% realised vol, the position profits $6.37 even before costs. At 45% realised vol, losses occur regardless—the fundamental risk of short gamma.

6.3 Portfolio Risk

The dashboard models multi-leg books with aggregated Greeks, spot sensitivity (±20% ladder), and historical VaR using 564 daily return scenarios since 2024. At 95% confidence: 1-day VaR = $147.44, CVaR = $293.55 (2× VaR confirms fat tails).

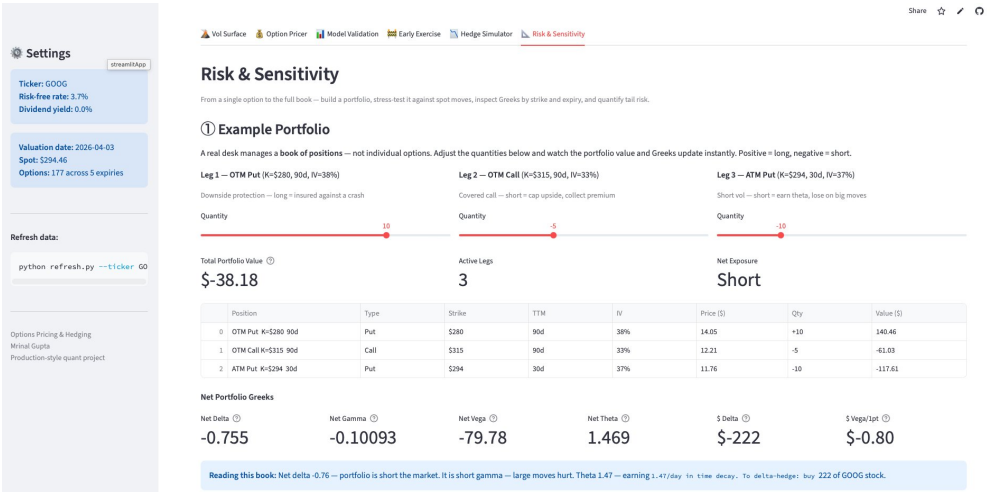

Figure 7: Portfolio risk dashboard—3-leg book with aggregated Greeks and hedge instructions.

7. Conclusions

This project demonstrates that a carefully constructed options pricing system can achieve production-quality accuracy on live market data. The iterative debugging process—14% to 94.4% hit rate across three market regimes—mirrors real quant work: identify failure, isolate root cause, fix, revalidate.

American IV extraction is the most impactful modelling choice. BSM inversion for American puts embeds early exercise premium into the volatility estimate, inflating put IV by 1–3 vol points. Switching to CRR inversion was the single largest improvement: 14% $\rightarrow$ 74.7% hit rate. This is a common practitioner mistake and the project demonstrates both the problem and the correct fix.

Surface calibration must be robust to stress. During the April 1 tariff selloff, OTM spreads widened to \$3–5, creating a 40:1 weight ratio that made wings invisible to the optimiser. ρ flipped to +0.46 (right skew—wrong for equity), collapsing hit rate to 12.1%. The minimum weight floor at 20% of ATM weight recovered to 81.1%—a simple fix preserving skew without surrendering to outliers.

Validation must use market-consistent metrics. Bid-ask hit rate tests consistency with tradeable levels, not just proximity to mid. BSM achieves low MAE by systematically underpricing puts while still failing the hit rate test. The 94.4% CRR result means the model produces prices a trader could execute against.

Hedging profitability depends on assumptions. Under ideal conditions, daily delta hedging removes 99.8% of variance. The stress matrix quantifies reality: at 20bp costs with realised vol 10pp above implied, the hedge loses \$7.42/contract. When realised vol is 10pp below implied, it profits \$4.42 after 20bp costs—the fundamental short gamma trade-off.

Limitations & Extensions

The system validates on GOOG only; multi-asset testing (SPY, AAPL, TSLA) would demonstrate generalisability. SVI preserves calendar no-arbitrage but does not enforce butterfly constraints—addressable via arbitrage-free SVI (Gatheral & Jacquier, 2014). Historical VaR uses 564 scenarios without regime scaling; filtered historical simulation would better capture non-stationary tails.

References [1] Black & Scholes (1973). *J. Political Economy*, 81(3), 637–654. [2] Cox, Ross & Rubinstein (1979). *J. Financial Economics*, 7(3), 229–263. [3] Longstaff & Schwartz (2001). *Rev. Financial Studies*, 14(1), 113–147. [4] Gatheral (2004). Global Derivatives, Madrid.